

1 Verilog Implementation

```
1 module phase4(  
2     input  clk,  
3     input  rst,  
4     input  [7:0] code_in,  
5     output reg phase4_done,  
6     output reg phase4_fail  
7 );  
8 parameter safe=3'b000;  
9 parameter s1=3'b001;  
10 parameter s2=3'b010;  
11 parameter s3=3'b011;  
12 parameter done=3'b101;  
13 parameter fail=3'b110;  
14 reg alaram;  
15 reg [2:0] current_state,next_state;  
16 always @(posedge clk) begin  
17     if(rst)  
18         current_state<=safe;  
19     else  
20         current_state<=next_state;  
21 end  
22 always @(*) begin  
23     phase4_done=0;  
24     phase4_fail=0;  
25     alaram=0;  
26     case(current_state)  
27         safe:  
28             next_state=s1;  
29         s1:  
30             if(code_in == 8'b10101010)  
31                 next_state=s2;  
32             else  
33                 next_state=fail;  
34         s2:  
35             if(code_in == 8'b11001100)  
36                 next_state=s3;  
37             else  
38                 next_state=fail;  
39         s3:  
40             if(code_in == 8'b11110000)  
41                 next_state=done;  
42             else  
43                 next_state=fail;  
44         done:  
45             phase4_done=1;  
46         fail: begin  
47             phase4_fail=1;  
48             alaram=1;  
49     end
```

```
50     default:
51         next_state=safe;
52     endcase
53 end
54 endmodule
```